

Qualification Pack



Solar Enterprise Assistant Manager

QP Code: SGJ/Q2601

Version: 1.0

NSQF Level: 5

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SGJ/Q2601: Solar Enterprise Assistant Manager

Brief Job Description

According to the Indian Renewable Energy Development Agency Limited (IREDA), India is endowed with abundant solar energy capable of producing 5,000 trillion kilowatts (kW) of clean energy. Moreover, India gets 300 sunny days a year in most parts of the country and solar insolation of 4-7 kWh per square meter per day. Strengthen domestic manufacturing of various technology components like solar cells, modules, batteries to generate indirect employment opportunities in these allied sectors. This course would help them to support industry players for implementation of projects across the country. To enhance their knowledge in the field of Solar Energy Technologies and its applications. The participants would be able to learn the knowledge on theory and practical session for quality aspect as qualitative and quantitative measurement. Course will cover both Theoretical as well as Practical sessions which help participants to understand the importance of solar energy applications.

Personal Attributes

This unit tells about Solar Enterprise

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N4066: Introduction to Renewable \(solar\)energy, climate change, tools used in solar energy, Electromagnetism](#)
2. [SGJ/N4067: Identify Components of solar Photovoltaic System](#)
3. [SGJ/N4068: Perform Site Assessment, Design, Sizing & Installation of Solar Home System Installation](#)
4. [SGJ/N4069: Perform Commissioning, Testing and O&M and Environmental Impacts of Solar PV system](#)
5. [SGJ/N4070: Understand Basics of Computers and E-Commerce](#)
6. [SGJ/N4071: Identify Geographic Information System of Solar PV system](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
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Sub-Sector	Renewable Energy
Occupation	Renewable Energy Professional
Country	India
NSQF Level	5
Credits	17
Aligned to NCO/ISCO/ISIC Code	2143.0100 (NCO, 2015) : Solar Energy System Designer 7421.1401/1402 (NCO, 2015) Solar Photo Voltaic System Installation Technician 7421.1403 (NCO, 2015) : PV System Installation Engineer
Minimum Educational Qualification & Experience	Diploma with NA of experience OR 12th grade Pass (2 year diploma in relevant field) with 1 Year of experience of relevant experience OR 10th grade pass (Plus 3 year diploma in relevant field) with 1 Year of experience of relevant experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience of relevant experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	21 Years
Last Reviewed On	NA
Next Review Date	31/01/2027
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-05-ES-01754-2023-V1-SCGJ
NQR Version	1.0

Remarks:

Total 510 (Theory: 120 hours: Practical: 180 hours: Employability Skills: 60 hours and OJT: 150 hours)

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SGJ/N4066: Introduction to Renewable (solar)energy, climate change, tools used in solar energy, Electromagnetism

Description

This unit tells about Introduction to Renewable (solar)energy, climate change, tools used in solar energy, Electromagnetism

Scope

The scope covers the following :

- Providing overview of Renewable energy and current information on renewable energy and related technologies.
- Explain Concept of Climate Change and about Solar Energy and its impacts on Climate Change .
- Learn about the use of tools and common myths about PPEs.
- Concepts of Electromagnetism and benefit from electromagnetic principles in the conversion of sunlight into heat energy.

Elements and Performance Criteria

Introduction to Renewable and Solar Energy fundamentals

To be competent, the user/individual on the job must be able to:

- PC1.** explain basics of renewable energy, Importance of Renewable energy, Types of renewable energy.
- PC2.** discuss status of Solar energy in India & Solar resource.
- PC3.** explain difference between renewable and non- renewable energy.
- PC4.** discuss permits and approvals, line diagram etc.
- PC5.** discuss about measurements and testing of system.
- PC6.** discuss about site risk , hazard assessment and safety management.
- PC7.** explain impact of solar energy in the economy- Green economy.

Solar Energy & Climate Change

To be competent, the user/individual on the job must be able to:

- PC8.** discuss differences of Fossil Fuel & Solar Energy.
- PC9.** explain effects of Solar energy with Earth's Atmosphere & Carbon Footprint.
- PC10.** discuss climate Change and Global Warming Action Plan for India.
- PC11.** discuss administrative Functions And Basics Of Enterprise Management cook stoves

Tools

To be competent, the user/individual on the job must be able to:

- PC12.** demonstrate about screw drivers, measuring tape, Wrench, Pliers, Hammers, Hacksaw, Cutters, Chisels, Allen Keys, Hand Drill & drill bit, Try Square, Gimlet, vice, Pipe cutter, Crimping tool, Neon tester etc.
- PC13.** show mallet, Wire stripper, Centre punch, Hand grinder etc.
- PC14.** demonstrate sessions on tools.

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PC15. ensure students can learn about the tools.

Electromagnetism (Electricity & Magnetism)

To be competent, the user/individual on the job must be able to:

PC16. explain brief Introduction to Electricity & Electrical circuits.

PC17. explain laws of Resistance, Ohms law's, Kirchhoff's law.

PC18. explain about instrument needed for Measuring Electric Power.

PC19. discuss magnetism, Magnetic Needle, Electric & Magnetic fields, Law of ampere, Solenoid, Self & Mutual Inductance, Hysteresis loss.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. understanding of the basic principles of solar energy capture and conversion.

KU2. knowledge of different solar energy technologies, such as photovoltaic (PV) systems, concentrated solar power (CSP), and solar thermal collectors.

KU3. ability to assess the suitability of different solar energy solutions for specific applications.

KU4. familiarity with government incentives and policies related to solar energy.

KU5. strong communication and interpersonal skills to explain solar energy concepts to clients and stakeholders.

KU6. understanding of the science of climate change, including the causes, impacts, and mitigation strategies.

KU7. ability to communicate complex scientific information to a variety of audiences.

KU8. knowledge of climate change policies and regulations.

KU9. understanding of the basic principles of electromagnetism, including electric fields, magnetic fields, and their interactions.

KU10. ability to apply electromagnetic principles to solve problems in solar energy, such as the design of solar cells and other components.

KU11. strong mathematical and analytical skills.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. design and installation of solar energy systems.

GS2. maintenance and repair of solar energy systems.

GS3. project management for solar energy projects.

GS4. financial analysis of solar energy projects.

GS5. climate change research and modeling.

GS6. development and implementation of climate change mitigation and adaptation strategies.

GS7. education and outreach on climate change issues.

GS8. design and development of solar cells and other electromagnetic devices.

GS9. research on new and innovative electromagnetic technologies for solar energy applications.



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GS10. teaching and research in electromagnetism and related fields.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Renewable and Solar Energy fundamentals</i>	35	-	-	-
PC1. explain basics of renewable energy, Importance of Renewable energy, Types of renewable energy.	5	-	-	-
PC2. discuss status of Solar energy in India & Solar resource.	5	-	-	-
PC3. explain difference between renewable and non- renewable energy.	5	-	-	-
PC4. discuss permits and approvals, line diagram etc.	5	-	-	-
PC5. discuss about measurements and testing of system.	5	-	-	-
PC6. discuss about site risk , hazard assessment and safety management.	5	-	-	-
PC7. explain impact of solar energy in the economy- Green economy.	5	-	-	-
<i>Solar Energy & Climate Change</i>	20	-	-	-
PC8. discuss differences of Fossil Fuel & Solar Energy.	5	-	-	-
PC9. explain effects of Solar energy with Earth's Atmosphere & Carbon Footprint.	5	-	-	-
PC10. discuss climate Change and Global Warming Action Plan for India.	5	-	-	-
PC11. discuss administrative Functions And Basics Of Enterprise Management cook stoves	5	-	-	-
<i>Tools</i>	-	25	-	-
PC12. demonstrate about screw drivers, measuring tape, Wrench, Pliers, Hammers, Hacksaw, Cutters, Chisels, Allen Keys, Hand Drill & drill bit, Try Square, Gimlet, vice, Pipe cutter, Crimping tool, Neon tester etc.	-	6	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. show mallet, Wire stripper, Centre punch, Hand grinder etc.	-	6	-	-
PC14. demonstrate sessions on tools.	-	7	-	-
PC15. ensure students can learn about the tools.	-	6	-	-
<i>Electromagnetism (Electricity & Magnetism)</i>	20	-	-	-
PC16. explain brief Introduction to Electricity & Electrical circuits.	5	-	-	-
PC17. explain laws of Resistance, Ohms law's, Kirchhoff's law.	5	-	-	-
PC18. explain about instrument needed for Measuring Electric Power.	5	-	-	-
PC19. discuss magnetism, Magnetic Needle, Electric & Magnetic fields, Law of ampere, Solenoid, Self & Mutual Inductance, Hysteresis loss.	5	-	-	-
NOS Total	75	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4066
NOS Name	Introduction to Renewable (solar)energy, climate change, tools used in solar energy, Electromagnetism
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Renewable Energy Professional
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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SGJ/N4067: Identify Components of solar Photovoltaic System

Description

This unit tells about Components of solar Photovoltaic System

Scope

The scope covers the following :

- Understanding of Electrical wires and cables.
- Identify Solar energy system components: Cell, Module, String, Array.
- Identify Components of Batteries, Inverters, Trackers.

Elements and Performance Criteria

Wires &Cables

To be competent, the user/individual on the job must be able to:

- PC1.** discuss about understanding electrical wire.
- PC2.** discuss about understanding electrical cable.
- PC3.** explain about insulating material, Standard wire gauge, Continuity tester, Soldering.
- PC4.** discuss safety and its importance, PPEs, Safety Signs, Safety Slogans, Safety Rules, Fire Extinguisher
- PC5.** perform practical sessions on wires and cables.

Photovoltaic Technology

To be competent, the user/individual on the job must be able to:

- PC6.** explain types of Solar cells (Crystalline Silicon, Thin film, Organic Photovoltaic cell/3rd generation PV cell).
- PC7.** explain Cell, Module, String, Array.
- PC8.** discuss STC & NOCT conditions.
- PC9.** discuss Solar cell I-V Characteristics Curves.
- PC10.** explain factors affect the performance of solar panel.
- PC11.** discuss difference between Solar PV and Solar Thermal Systems.
- PC12.** demonstration of Solar PV and its components.
- PC13.** demonstrate solar cell characteristics.

Components of Photovoltaic System

To be competent, the user/individual on the job must be able to:

- PC14.** discuss about solar panels.
- PC15.** explain mechanism of Batteries, Inverter, Charge Controllers.
- PC16.** explain mechanism of trackers, balance of system.
- PC17.** discuss monitoring & metering system.
- PC18.** demonstration of Batteries, Inverters, Charge Controllers etc.
- PC19.** perform hands on training on batteries, inverters etc.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** understanding the principles of solar energy, including how sunlight is converted into electricity using photovoltaic cells, is fundamental to grasping the purpose and function of the components in a solar PV system.
- KU2.** knowledge of the different types of photovoltaic cells (e.g., monocrystalline, polycrystalline, thin-film) and how they are assembled into solar panels/modules is essential.
- KU3.** understanding the types of inverters (string, micro, central), their efficiency, and their role in system performance is important.
- KU4.** inverters are crucial for converting the direct current (DC) electricity generated by solar panels into alternating current (AC) electricity that can be used by household appliances and the electric grid.
- KU5.** understanding the electrical wiring, connectors, junction boxes, combiner boxes, and other electrical components is necessary for designing and installing a safe and efficient solar PV system.
- KU6.** knowledge of local, national, and international regulations and safety standards related to solar PV systems is important for compliance and ensuring the safety and reliability of the system.
- KU7.** understanding the principles of system design, including site assessment, sizing, shading analysis, and orientation, is important for designing an efficient and effective solar PV system.
- KU8.** knowledge of installation techniques, best practices, and quality assurance is also essential.
- KU9.** knowledge of the economic and environmental benefits of solar PV systems, including cost savings, energy independence, and reduced carbon emissions, is important for decision-making and advocacy.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** solar PV systems are complex, so it is important to be able to read and understand technical drawings and specifications.
- GS2.** understanding how solar cells work and how they convert sunlight into electricity.
- GS3.** staying informed about the latest advancements in solar PV technology, such as bifacial panels, smart inverters, and energy storage solutions, is important for staying competitive and offering innovative solutions to customers.
- GS4.** knowledge of direct current (DC) and alternating current (AC) electricity, voltage, amperage, resistance, and power.
- GS5.** solar panels are often installed on roofs, so workers need to be able to climb ladders and work at heights.
- GS6.** ability to work safety protocols such as wearing fall protection, using electrical safety equipment, and being aware of potential hazards.
- GS7.** solar PV systems are complex, so it is important to be able to read and understand technical drawings and specifications.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Wires & Cables</i>	20	5	-	-
PC1. discuss about understanding electrical wire.	5	-	-	-
PC2. discuss about understanding electrical cable.	5	-	-	-
PC3. explain about insulating material, Standard wire gauge, Continuity tester, Soldering.	5	-	-	-
PC4. discuss safety and its importance, PPEs, Safety Signs, Safety Slogans, Safety Rules, Fire Extinguisher	5	-	-	-
PC5. perform practical sessions on wires and cables.	-	5	-	-
<i>Photovoltaic Technology</i>	30	11	-	-
PC6. explain types of Solar cells (Crystalline Silicon, Thin film, Organic Photovoltaic cell/3rd generation PV cell).	5	-	-	-
PC7. explain Cell, Module, String, Array.	5	-	-	-
PC8. discuss STC & NOCT conditions.	5	-	-	-
PC9. discuss Solar cell I-V Characteristics Curves.	5	-	-	-
PC10. explain factors affect the performance of solar panel.	5	-	-	-
PC11. discuss difference between Solar PV and Solar Thermal Systems.	5	-	-	-
PC12. demonstration of Solar PV and its components.	-	6	-	-
PC13. demonstrate solar cell characteristics.	-	5	-	-
<i>Components of Photovoltaic System</i>	20	14	-	-
PC14. discuss about solar panels.	5	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. explain mechanism of Batteries, Inverter, Charge Controllers.	5	-	-	-
PC16. explain mechanism of trackers, balance of system.	5	-	-	-
PC17. discuss monitoring & metering system.	5	-	-	-
PC18. demonstration of Batteries, Inverters, Charge Controllers etc.	-	7	-	-
PC19. perform hands on training on batteries, inverters etc.	-	7	-	-
NOS Total	70	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4067
NOS Name	Identify Components of solar Photovoltaic System
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Renewable Energy Professional
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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SGJ/N4068: Perform Site Assessment, Design, Sizing & Installation of Solar Home System Installation

Description

This unit tells about Perform Site Assessment, Design, Sizing & Installation of Solar Home System Installation

Scope

The scope covers the following :

- Acquire Advanced knowledge within the field of photovoltaic system technology and study of solar photovoltaic (PV) cells, modules, and system components, electrical circuits, PV system etc.
- Perform PV installation and requirements and Cost estimation of the system.
- Perform Evaluation of site suitability, solar access, shadowing considerations, site approach, power evacuation and other variables.

Elements and Performance Criteria

Types Of PV Systems

To be competent, the user/individual on the job must be able to:

- PC1.** discuss about PV Direct System, Off Grid System.
- PC2.** explain grid connected systems (with & without battery).
- PC3.** explain hybrid system.
- PC4.** explain photovoltaic mounting systems.
- PC5.** explain roof mounted system.
- PC6.** explain ground mounted system.
- PC7.** explain building Integrated photovoltaic.
- PC8.** discuss mounting as a shade structure.
- PC9.** demonstration of PV system.
- PC10.** show off-grid and Grid connected.
- PC11.** demonstration of roof mounted system.

PV System Design, Sizing & Installation Requirements

To be competent, the user/individual on the job must be able to:

- PC12.** determine power consumption demands.
- PC13.** discuss about size of the PV panels.
- PC14.** explain battery, inverter, charge controller sizing's.
- PC15.** discuss about sizing system wiring.
- PC16.** discuss about solar PV system installation requirements.
- PC17.** discuss about permits & approvals for PV installation design & installation checklist.
- PC18.** discuss various costs involved in solar PV project.
- PC19.** perform designing of solar PV system.

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PC20. perform PV installation and requirements.

PC21. draw line diagram for installation.

Site Assessment for Solar Home System Installation

To be competent, the user/individual on the job must be able to:

PC22. perform site assessment at field.

PC23. perform installation of solar module.

PC24. perform installation of charge controller/regulator.

PC25. perform installation of battery.

PC26. perform wiring of solar home system components.

PC27. show lamp, switch & power socket installation procedures.

PC28. show components assembly of solar home systems.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. understand roof orientation, dimensions, pitch, strength, and shading from trees, buildings, etc.

KU2. know zoning permits, net metering policies, and interconnection requirements.

KU3. analyze historical electricity bills to estimate system size needed.

KU4. understand different types of solar panels, inverters, batteries, and mounting systems.

KU5. knowledge system cost, potential energy savings, rebates, and payback period.

KU6. securely mount panels on the roof using appropriate racking systems.

KU7. knowledge testing and verify system performance and functionality.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. understanding of solar panels work, how to calculate solar energy production, and the different types of solar systems available.

GS2. understanding AC and DC electricity, electrical safety procedures, and how to connect solar panels to an electrical grid.

GS3. there are many factors to consider when designing and installing a solar system, so being able to identify and solve problems is essential.

GS4. able to communicate effectively with customers, answer their questions, and address their concerns.

GS5. ability to use software programs available that can help with solar system design and sizing. Being able to use these programs is a valuable skill.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Types Of PV Systems</i>	24	15	-	-
PC1. discuss about PV Direct System, Off Grid System.	3	-	-	-
PC2. explain grid connected systems (with & without battery).	3	-	-	-
PC3. explain hybrid system.	3	-	-	-
PC4. explain photovoltaic mounting systems.	3	-	-	-
PC5. explain roof mounted system.	3	-	-	-
PC6. explain ground mounted system.	3	-	-	-
PC7. explain building Integrated photovoltaic.	3	-	-	-
PC8. discuss mounting as a shade structure.	3	-	-	-
PC9. demonstration of PV system.	-	5	-	-
PC10. show off-grid and Grid connected.	-	5	-	-
PC11. demonstration of roof mounted system.	-	5	-	-
<i>PV System Design, Sizing & Installation Requirements</i>	26	15	-	-
PC12. determine power consumption demands.	4	-	-	-
PC13. discuss about size of the PV panels.	3	-	-	-
PC14. explain battery, inverter, charge controller sizing's.	3	-	-	-
PC15. discuss about sizing system wiring.	4	-	-	-
PC16. discuss about solar PV system installation requirements.	4	-	-	-
PC17. discuss about permits & approvals for PV installation design & installation checklist.	4	-	-	-
PC18. discuss various costs involved in solar PV project.	4	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC19. perform designing of solar PV system.	-	5	-	-
PC20. perform PV installation and requirements.	-	5	-	-
PC21. draw line diagram for installation.	-	5	-	-
<i>Site Assessment for Solar Home System Installation</i>	-	20	-	-
PC22. perform site assessment at field.	-	2	-	-
PC23. perform installation of solar module.	-	2	-	-
PC24. perform installation of charge controller/regulator.	-	3	-	-
PC25. perform installation of battery.	-	3	-	-
PC26. perform wiring of solar home system components.	-	3	-	-
PC27. show lamp, switch & power socket installation procedures.	-	3	-	-
PC28. show components assembly of solar home systems.	-	4	-	-
NOS Total	50	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4068
NOS Name	Perform Site Assessment, Design, Sizing & Installation of Solar Home System Installation
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Renewable Energy Professional
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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SGJ/N4069: Perform Commissioning, Testing and O&M and Environmental Impacts of Solar PV system

Description

This unit tells about Perform Commissioning, Testing and O&M and Environmental Impacts of Solar PV system

Scope

The scope covers the following :

- Learn about Testing, commissioning, Operations and Maintenance of PV systems.
- Explain Scenario of environmental impact on solar PV.
- Perform Recycling for Solar PV and Waste of Electrical & Electronic Equipment.

Elements and Performance Criteria

Commissioning, Testing and O&M of PV Systems

To be competent, the user/individual on the job must be able to:

- PC1.** discuss final installation checkout.
- PC2.** discuss about visual inspection.
- PC3.** discuss about system testing & measurements.
- PC4.** explain about testing PV system.
- PC5.** discuss about operation & maintenance of the system.
- PC6.** discuss about troubleshooting of PV system & other common issues.
- PC7.** discuss about site risk & hazard assessment.
- PC8.** perform commissioning and testing of system.
- PC9.** perform hands on experience.
- PC10.** perform troubleshooting of PV system.

Environmental Impacts of Solar PV system

To be competent, the user/individual on the job must be able to:

- PC11.** discuss about sustainable source of energy.
- PC12.** explain about stable energy prices.
- PC13.** discuss about reliable energy system.
- PC14.** discuss about land use, water use & natural resource use.
- PC15.** explain life cycle global warming emissions.
- PC16.** discuss about job and other economic benefits.
- PC17.** explain hazardous materials & effects of E-Waste on human body.

Recycling of WEEE

To be competent, the user/individual on the job must be able to:

- PC18.** explain basics of recycling for solar PV.

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- PC19.** discuss about lack of regulations, recycling of solar cells, global experience, indian manufacturers, future of recycling.
- PC20.** discuss about recycle technology of solar panels & batteries.
- PC21.** discuss about waste electrical & electronic equipment.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** understanding of commissioning and verifying proper installation, testing functionality of individual components and the entire system, ensuring compliance with standards and regulations, and documenting results.
- KU2.** knowledge of different testing procedures like open-circuit voltage (OCV), short-circuit current (Isc), insulation resistance, ground fault testing, and performance ratio testing.
- KU3.** understanding of inverter, setting parameters, and interpreting data from monitoring systems to assess system performance.
- KU4.** understanding of common issues like inverter faults, module degradation, and shading, and knowing how to address them.
- KU5.** understanding the environmental impact of manufacturing, installation, operation, and decommissioning of PV systems.
- KU6.** knowledge of local environmental regulations and permits related to land use, waste management, and potential environmental hazards.
- KU7.** understanding the contribution of solar energy to reducing air pollution, mitigating climate change, and promoting renewable energy sources.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** ability to operating instruments like multimeters, clamp meters, insulation testers, and data loggers for accurate measurements.
- GS2.** analyzing system performance data to identify trends, diagnose potential issues, and optimize energy production.
- GS3.** knowing the terms and conditions for maintenance services and warranty coverage on system components.
- GS4.** implementing measures like recycling materials, minimizing water usage, and reducing greenhouse gas emissions during installation and operation.
- GS5.** interpreting test results and pinpointing performance issues are crucial for effective commissioning and O&M.
- GS6.** using tools like PVsyst, Helioscope, and PV F-Chart for system design and performance analysis.
- GS7.** staying updated on new technologies, regulations, and best practices.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Commissioning, Testing and O&M of PV Systems</i>	14	10	-	-
PC1. discuss final installation checkout.	2	-	-	-
PC2. discuss about visual inspection.	2	-	-	-
PC3. discuss about system testing & measurements.	2	-	-	-
PC4. explain about testing PV system.	2	-	-	-
PC5. discuss about operation & maintenance of the system.	2	-	-	-
PC6. discuss about troubleshooting of PV system & other common issues.	2	-	-	-
PC7. discuss about site risk & hazard assessment.	2	-	-	-
PC8. perform commissioning and testing of system.	-	3	-	-
PC9. perform hands on experience.	-	3	-	-
PC10. perform troubleshooting of PV system.	-	4	-	-
<i>Environmental Impacts of Solar PV system</i>	18	-	-	-
PC11. discuss about sustainable source of energy.	2	-	-	-
PC12. explain about stable energy prices.	2	-	-	-
PC13. discuss about reliable energy system.	2	-	-	-
PC14. discuss about land use, water use & natural resource use.	3	-	-	-
PC15. explain life cycle global warming emissions.	3	-	-	-
PC16. discuss about job and other economic benefits.	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. explain hazardous materials & effects of E-Waste on human body.	3	-	-	-
<i>Recycling of WEEE</i>	8	-	-	-
PC18. explain basics of recycling for solar PV.	2	-	-	-
PC19. discuss about lack of regulations, recycling of solar cells, global experience, indian manufacturers, future of recycling.	2	-	-	-
PC20. discuss about recycle technology of solar panels & batteries.	2	-	-	-
PC21. discuss about waste electrical & electronic equipment.	2	-	-	-
NOS Total	40	10	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4069
NOS Name	Perform Commissioning, Testing and O&M and Environmental Impacts of Solar PV system
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Renewable Energy Professional
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4070: Understand Basics of Computers and E-Commerce

Description

This unit tells about Basics of Computers and E-Commerce

Scope

The scope covers the following :

- Explain functions of computer and get hands-on training on MS-Word, Excel and payment gateways and Basics of E-Commerce.

Elements and Performance Criteria

Basics of Computers Understanding E-Commerce

To be competent, the user/individual on the job must be able to:

- PC1.** explain what is computer.
- PC2.** explain components of computer.
- PC3.** explain hardware & software.
- PC4.** explain basics of e-commerce.
- PC5.** show basics of ms-word.
- PC6.** show basics of MS-Excel.
- PC7.** demonstrate about payment gateways.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** knowledge basic components like, CPU (central processing unit), RAM (random access memory), storage (hard drive, SSD), motherboard, graphics card, network card, input devices (keyboard, mouse), output devices (monitor, printer).
- KU2.** knowledge how to operate Windows, macOS, Linux (different functionalities, interfaces, and purposes).
- KU3.** understanding of applications for various tasks like, (word processing, spreadsheets, web browsing, gaming, etc.).
- KU4.** ability to use a computer, navigate the internet, and use common software applications such as Microsoft Office.
- KU5.** ability to managing and organizing emails, using email etiquette, and using email to communicate effectively with colleagues, clients, and customers.
- KU6.** basic knowledge of buying and selling goods and services online using electronic transactions.
- KU7.** knowledge of platforms for browsing, selecting, and purchasing products.
- KU8.** understanding how to optimize your website for search engines is essential for driving traffic to your online store.

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- KU9.** understanding how to manage databases is essential for many jobs, especially those that deal with large amounts of data.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** understanding of how computers work, including basic hardware and software knowledge, is essential.
- GS2.** knowing how computer networks function and how to troubleshoot basic issues is valuable.
- GS3.** ability to collect, organize, and analyze data is crucial for making informed decisions in e-commerce.
- GS4.** being able to identify problems, analyze their causes, and develop effective solutions is a must-have skill in any field, especially e-commerce.
- GS5.** understanding how to market products and services online through channels like search engines, social media, and email marketing is essential for e-commerce success.
- GS6.** knowing how to convince customers to buy your products or services is a key sales skill.
- GS7.** being able to organize your time and resources effectively is crucial for completing projects on time and within budget.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basics of Computers Understanding E-Commerce</i>	20	30	-	-
PC1. explain what is computer.	5	-	-	-
PC2. explain components of computer.	5	-	-	-
PC3. explain hardware & software.	5	-	-	-
PC4. explain basics of e-commerce.	5	-	-	-
PC5. show basics of ms-word.	-	10	-	-
PC6. show basics of MS-Excel.	-	10	-	-
PC7. demonstrate about payment gateways.	-	10	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4070
NOS Name	Understand Basics of Computers and E-Commerce
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Renewable Energy Professional
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N4071: Identify Geographic Information System of Solar PV system

Description

This unit tells about Identify Geographic Information System of Solar PV system

Scope

The scope covers the following :

- Understand the concept of GIS mapping and importance of GIS mapping.

Elements and Performance Criteria

Geographic Information System (GIS)

To be competent, the user/individual on the job must be able to:

- PC1.** explain what is GIS.
- PC2.** discuss about tools used to store, visualize, analyse, and interpret geographic data.
- PC3.** ensure practical understanding of GIS mapping.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** basic knowledge of GIS concepts and tools like, spatial data analysis, mapping, and geoprocessing.
- KU2.** understanding of solar energy principles solar radiation, panel efficiency, and system design.
- KU3.** familiarity with relevant solar radiation data, land cover maps, and infrastructure datasets.
- KU4.** identify and mitigate shading from trees, buildings, or other structures.
- KU5.** optimize panel tilt angle, orientation, and spacing for maximum energy generation.
- KU6.** analyze compatibility with existing power lines and distribution networks.
- KU7.** analyze factors like solar radiation, land cover, slope, aspect, and proximity to power lines to find optimal sites for solar farms or rooftop installations.
- KU8.** ensure projects comply with zoning regulations and setback requirements.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** understanding the components, design considerations, and performance characteristics of solar PV systems is essential for applying GIS effectively in this field.
- GS2.** experience with geospatial data acquisition, processing, manipulation, and visualization.
- GS3.** expertise in using popular GIS software packages like ArcGIS, QGIS, or Erdas Imagine is highly sought-after.

Qualification Pack

- GS4.** several specialized GIS tools are available for solar energy analysis, such as Sun GIS, Heliodon, and PVsyst.
- GS5.** knowledge of permitting requirements, environmental regulations, and building codes related to solar PV systems is valuable.
- GS6.** ability to manage projects effectively and communicate with stakeholders is important for successful implementation of solar PV systems.
- GS7.** ability to interpret aerial imagery and LiDAR data can be beneficial for site assessments and resource mapping.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Geographic Information System (GIS)</i>	25	25	-	-
PC1. explain what is GIS.	15	-	-	-
PC2. discuss about tools used to store, visualize, analyse, and interpret geographic data.	10	-	-	-
PC3. ensure practical understanding of GIS mapping.	-	25	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4071
NOS Name	Identify Geographic Information System of Solar PV system
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Renewable Energy Professional
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council/Awarding Body. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC/AB will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC/AB.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4066.Introduction to Renewable (solar)energy, climate change, tools used in solar energy, Electromagnetism	75	25	0	0	100	20
SGJ/N4067.Identify Components of solar Photovoltaic System	70	30	0	0	100	20
SGJ/N4068.Perform Site Assessment, Design, Sizing & Installation of Solar Home System Installation	50	50	0	0	100	20
SGJ/N4069.Perform Commissioning, Testing and O&M and Environmental Impacts of Solar PV system	40	10	0	0	50	10
SGJ/N4070.Understand Basics of Computers and E-Commerce	20	30	0	0	50	10
SGJ/N4071.Identify Geographic Information System of Solar PV system	25	25	0	0	50	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	300	200	0	0	500	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.